

The empirical recurrence for OEIS A251098, recovered from its tail

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Abstract

OEIS A251098 records “Empirical recurrence of order 57” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 128$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 59$. Its last coefficient is $c_{57} = 679477248 \neq 0$, so no further tail satisfies anything shorter; any order-57 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A251098 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with no 2×2 subblock having its minimum diagonal element less than its minimum antidiagonal element.”. It has offset 1 and begins

4023, 160086, 6151051, 237288487, 9160138107, 353489877714, 13640572890821, 526395665662695, ...

The entry states:

Empirical recurrence of order 57 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:04:03, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 128$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 128$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 57: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 57.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 256$ exact terms from $n = 2$ on, it returns order 57 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	128	c_{16}	−478987657981661687	c_{31}	13378819240871794419863	c_{46}	−84596740
c_2	−7416	c_{17}	1708535027396077459	c_{32}	−14351999139655152172823	c_{47}	24858076
c_3	268855	c_{18}	−5537883443882043309	c_{33}	14293652520325723771702	c_{48}	−6512255
c_4	−6962742	c_{19}	16372356344425115187	c_{34}	−13207294330053725447653	c_{49}	15081283
c_5	138212489	c_{20}	−44298166214680657073	c_{35}	11311842691912039401532	c_{50}	−3053339
c_6	−2194860484	c_{21}	110018719608218142783	c_{36}	−8970481342198408381640	c_{51}	5325656
c_7	28687902493	c_{22}	−251475896247476436415	c_{37}	6577784342030406007808	c_{52}	−784542
c_8	−314916023388	c_{23}	530230419356568967277	c_{38}	−4452949801778032891252	c_{53}	949458
c_9	2947807689091	c_{24}	−1033264880914352655214	c_{39}	2778090090193673202072	c_{54}	−90639
c_{10}	−23814068541590	c_{25}	1863946925411975624314	c_{40}	−1594050437509513177264	c_{55}	63992
c_{11}	167685136809878	c_{26}	−3116656137264415380733	c_{41}	839337924161758010800	c_{56}	−2969
c_{12}	−1037831046570966	c_{27}	4835115444210724773947	c_{42}	−404534977680721185552	c_{57}	6794
c_{13}	5687092874649547	c_{28}	−6964590593285182927658	c_{43}	177962430375806198592		
c_{14}	−27769547778404699	c_{29}	9318597162387223086749	c_{44}	−71228172050131367808		
c_{15}	121517724327644086	c_{30}	−11584076980584371157197	c_{45}	25840565293904648576		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 59$, and it is the recurrence OEIS A251098 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{57} - c_1 x^{56} - \dots - c_{57}$. Its constant term is $-c_{57} = -679477248$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 57 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 57, so $\deg q = 57$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 57, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 13 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 57, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 679477248.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-57 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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